

Design an experiment to observe process state transitions (Ready, Running, Waiting, Terminated) using Linux monitoring tools such as PS, top, and /proc. Document the observations.

Code:

```
geya@LenovoL480: ~/OS/Pra x + v
GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>

int main() {
    char command[100];

    printf("Enter Linux command: ");
    fgets(command, sizeof(command), stdin);

    printf("\nExecuting command...\n");
    system(command);

    return 0;
}
```

Output:

```
geya@LenovoL480:~/OS/Practical/Week 3$ ./task2
Enter Linux command: ps

Executing command...
  PID TTY          TIME CMD
  325 pts/0        00:00:00 bash
 2890 pts/0        00:00:00 task2
 2891 pts/0        00:00:00 sh
 2892 pts/0        00:00:00 ps
```

```
geya@LenovoL480:~/OS/Practical/Week 3$ ./task2
Enter Linux command: top
```

Executing command...

```
top - 05:27:20 up 1:19, 1 user, load average: 0.05, 0.01, 0.00
Tasks: 26 total, 1 running, 25 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 15931.6 total, 15328.0 free, 573.2 used, 227.6 buff/cache
MiB Swap: 4096.0 total, 4096.0 free, 0.0 used, 15358.4 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	24144	15388	11576	S	0.0	0.1	0:01.56	systemd
2	root	20	0	3180	2208	2072	S	0.0	0.0	0:00.02	init-systemd(Ub
7	root	20	0	3180	2044	1940	S	0.0	0.0	0:00.00	init
47	root	19	-1	50400	16832	15512	S	0.0	0.1	0:00.50	systemd-journal
82	systemd+	20	0	22540	14532	11916	S	0.0	0.1	0:00.11	systemd-resolve
88	root	20	0	35248	12388	9244	S	0.0	0.1	0:21.50	systemd-udev
136	root	20	0	2888	1948	1816	S	0.0	0.0	0:00.07	chronyd-starter
141	root	20	0	4472	3088	2844	S	0.0	0.0	0:00.01	cron
144	message+	20	0	8816	5388	4716	S	0.0	0.0	0:00.12	dbus-daemon
154	root	20	0	44092	29712	14576	S	0.0	0.2	0:00.34	networkd-dispat
163	root	20	0	18440	9400	8152	S	0.0	0.1	0:00.19	systemd-logind
185	syslog	20	0	220548	5948	4656	S	0.0	0.0	0:00.13	rsyslogd
221	_chrony	20	0	24440	11960	7172	S	0.0	0.1	0:00.33	chronyd
233	_chrony	20	0	12092	2404	1760	S	0.0	0.0	0:00.02	chronyd
248	root	20	0	5492	2696	2508	S	0.0	0.0	0:00.02	agetty
249	root	20	0	123456	32720	18108	S	0.0	0.2	0:00.26	unattended-upgr
318	root	20	0	3188	1116	980	S	0.0	0.0	0:00.00	SessionLeader
320	root	20	0	3204	1256	1108	S	0.0	0.0	0:00.43	Relay(325)
325	geya	20	0	6268	5520	3848	S	0.0	0.0	0:00.41	bash

```
geya@LenovoL480:~/OS/Practical/Week 3$ ./task2
Enter Linux command: ls /proc
```

Executing command...

```
1 233 320 88 crypto fs kmsg modules scsi thread-self
136 248 325 ACPI devices interrupts kpagecgroup mounts self timer_list
141 249 328 buddyinfo diskstats iomem kpagecount mpt slabinfo tty
144 3002 380 bus dma ioports kpageflags mtrr softirqs uptime
154 3003 382 cgroups driver irq latency_stats net stat version
163 3004 411 cmdline dynamic_debug kallsyms loadavg pagetypeinfo swaps vmallocinfo
185 3005 47 config.gz execdomains kcore locks partitions sys vmstat
2 3006 7 consoles fb key-users meminfo pressure sysrq-trigger zoneinfo
221 318 82 cpuinfo filesystems keys misc schedstat sysvipc
geya@LenovoL480:~/OS/Practical/Week 3$
```